* $x_1, \dots, x_n$ Poisson'a uysun MLE bulalım

$$f(x_0 | \lambda) = \frac{\lambda^{x_0} \cdot e^{-\lambda}}{x_0!} \rightarrow x_0 \text{ olasılığı}$$

$$L(\theta) = f(x_1, \dots, x_n | \theta) = f(x_2, \dots, x_n | x_1, \theta) \cdot f(x_3, \dots, x_n | x_1, x_2, \theta) \dots f(x_n | x_1, \dots, x_{n-1}, \theta)$$

$\downarrow$ Bağımsızlık varsayımı

$$L(\theta) = f(x_1 | \theta) \cdot f(x_2 | \theta) \dots f(x_n | \theta) = \prod_{i=1}^{n} \frac{\theta^{x_i} \cdot e^{-\theta}}{x_i!}$$

$\downarrow$

$$\ln L(\theta) = \sum_{i=1}^{n} \ln \frac{\theta^{x_i} \cdot e^{-\theta}}{x_i!} = \sum_{i=1}^{n} x_i \cdot \ln\theta - \theta \cdot \overset{1}{\ln e} - [\ln(x_i) + \ln(x_i - 1) + \dots + \ln(x_i - x_i + 1)]$$

$$\frac{\partial \ln L(\theta)}{\partial \theta} = \sum_{i=1}^{n} \frac{x_i}{\theta} - 1 - [0 + 0 + \dots + 0] = \frac{1}{\theta} \sum_{i=1}^{n} [x_i] - n = 0$$ local bulcaz

$$\frac{\partial \ln L(\theta)}{\partial \theta} = \frac{1}{\theta} \sum_{i=1}^{n} [x_i] - n$$

	$\bar{x}$	
$\frac{\partial \ln L(\theta)}{\partial \theta}$	+	-

$\theta <$ mean ise

$\downarrow$

$\frac{1}{\theta} \cdot x_{sum} > n$

$\theta >$ mean ise

$\downarrow$

$\frac{1}{\theta} \cdot x_{sum} < n$

$$\frac{1}{\theta} \cdot \sum_{i=1}^{n} x_i = n$$

$\downarrow$

$$\theta = \frac{1}{n} \cdot \sum_{i=1}^{n} x_i$$ $\rightarrow \theta$ sample mean çıkıyor

Veya Hessian analizi de olurdu